

ANALOG ELECTRONICS

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SUMMARY

- Session 1:
  - How to use MATLAB and LTspice
  - MOS design curves and trade-offs
  - Common source amplifier with resistive load
  - Common source amplifier with PMOS load

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THE TA TEAM



Arno Hemelhof
Master's degree from VUB/BRUFACE (Belgium, 2022)
Works on designing robust and efficient mmWave power amplifiers for 6G applications
Speaks Dutch and English



Chang-Shao Shen
Master's degree from NTU (Taiwan, 2021)
Works on electronic and photonic circuit design of distributed radar transceivers
Speaks Chinese and English



Ezgi Saç
Master's degree from KULeuven (Belgium, 2024)
Works on the design of digital transmit front-end in III-V technology for 5/6G applications
Speaks Turkish and English

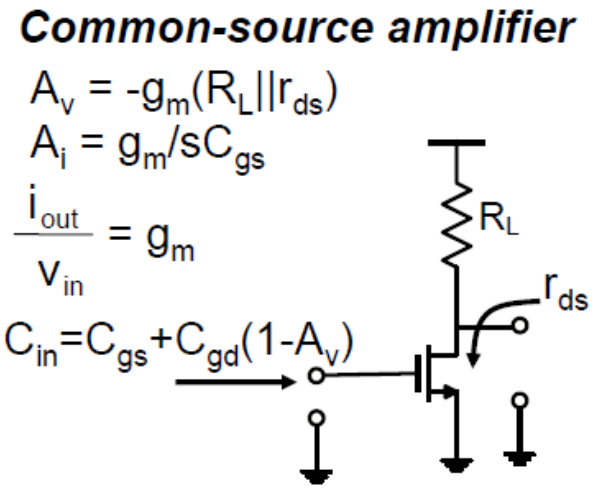


Xuepu Wu
Master's degree from ZJU (China, 2023)
Works on complex-sensing integrated circuits for built-in self-test of millimeter-wave transceivers
Speaks Chinese and English

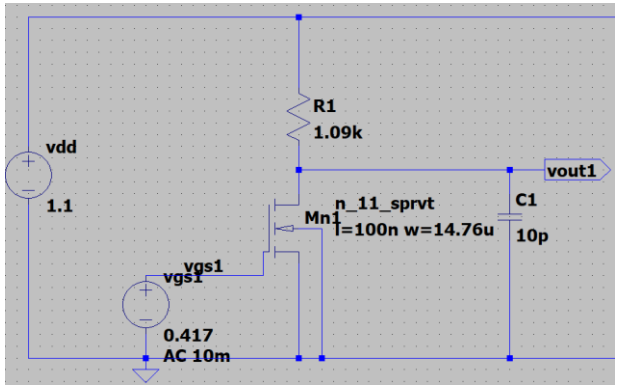
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CORE DIFFERENCES BETWEEN LECTURES AND LAB CLASSES

- Lectures:
  - Analysis of given architectures
  - Modeling of transistors and complex circuits
  - Finding equations for performance parameters (gain, bandwidth, power,...)



- Lab:
  - Quantitative design of the known architectures
  - How to size a transistor (width and length) to achieve a certain performance using Matlab
  - How to simulate a circuit using SPICE to validate the correctness of design



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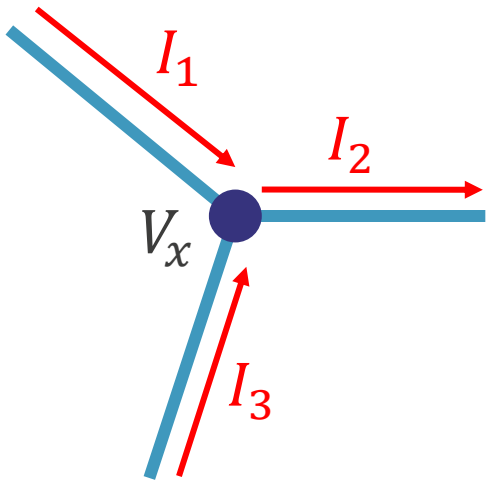
CONVENTION

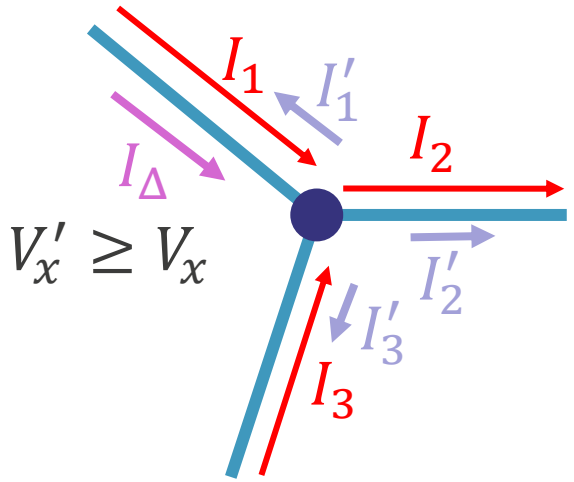
- DC: all uppercase (I_{DS}, V_{GS}, ...)
- Small-signal: all lowercase (i_{ds}, v_{gs}, ...)
- Large-signal: mixed case (I_{ds}, V_{gs}, ...)

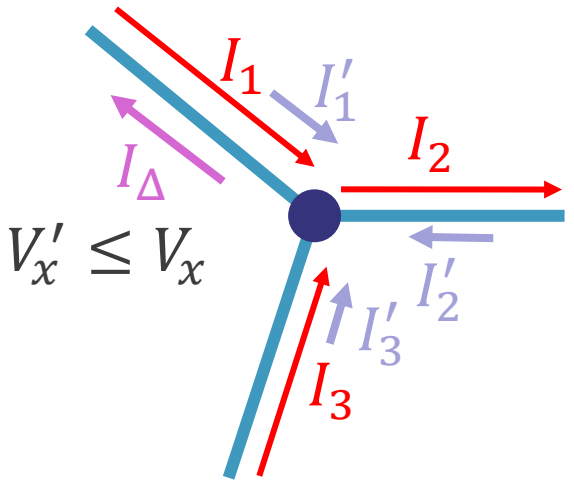
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PRIMER
REASONING ABOUT CIRCUITS

Kirchhoff's Current Law (KCL): $I_1 - I_2 + I_3 = 0$







Injecting current into a node increases the voltage

Extracting current from a node decreases the voltage

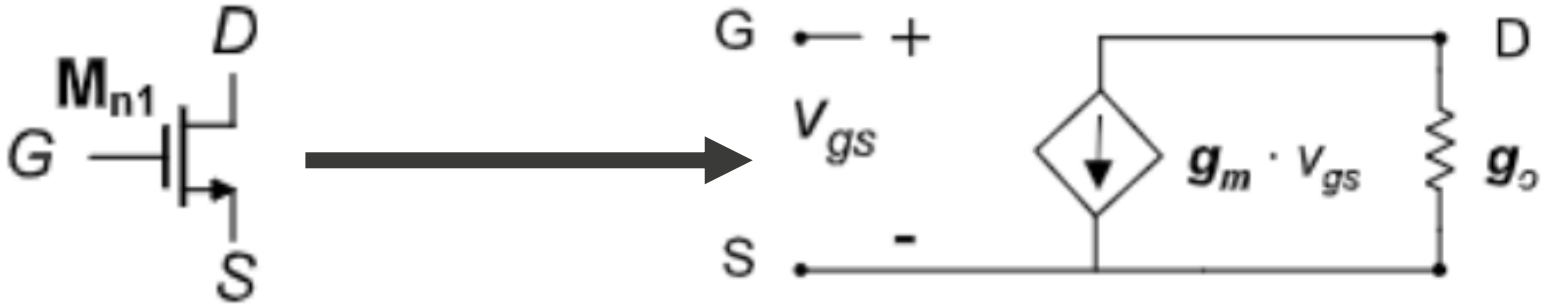
$$|\Delta V_x| = I_\Delta \cdot R_x$$

R_x is the equivalent resistance seen at node x
(including the branch the branch containing I_Δ)

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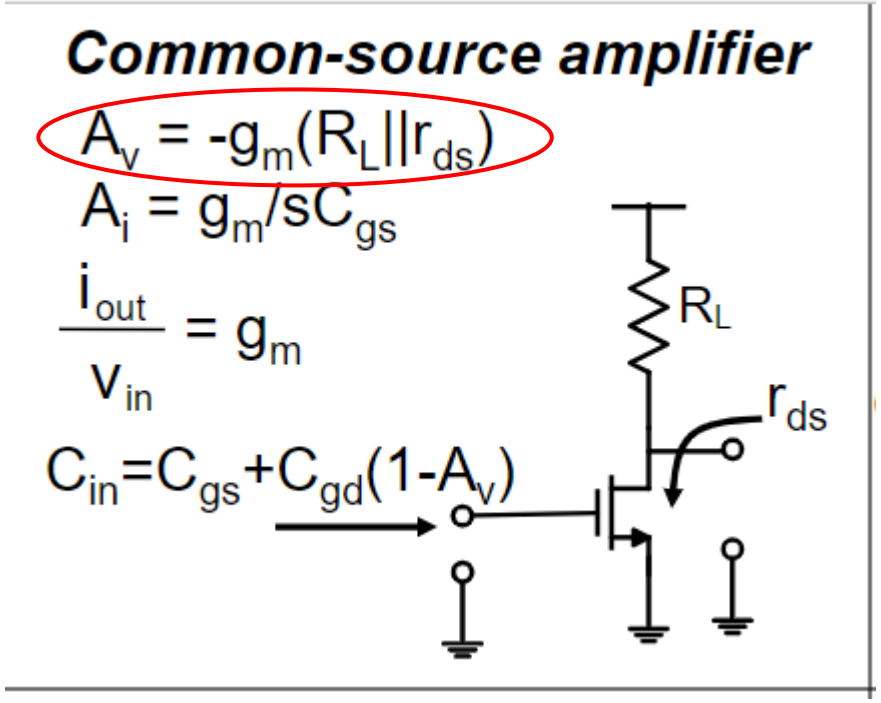
PRIMER

REASONING ABOUT CIRCUITS



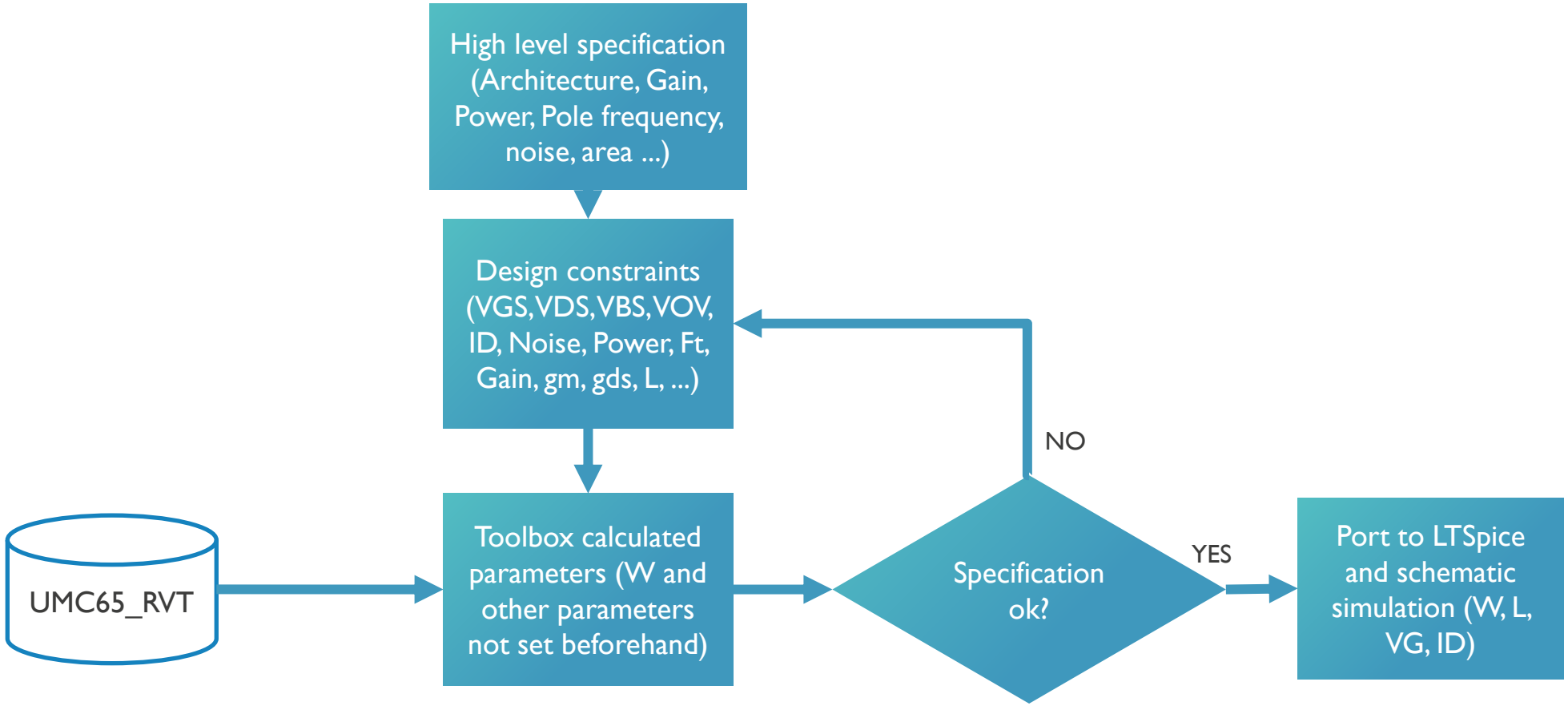
Also knows as g_{ds}

Increasing v_{gs} will increase i_{ds}, injecting current into **S** and extracting current from **D**



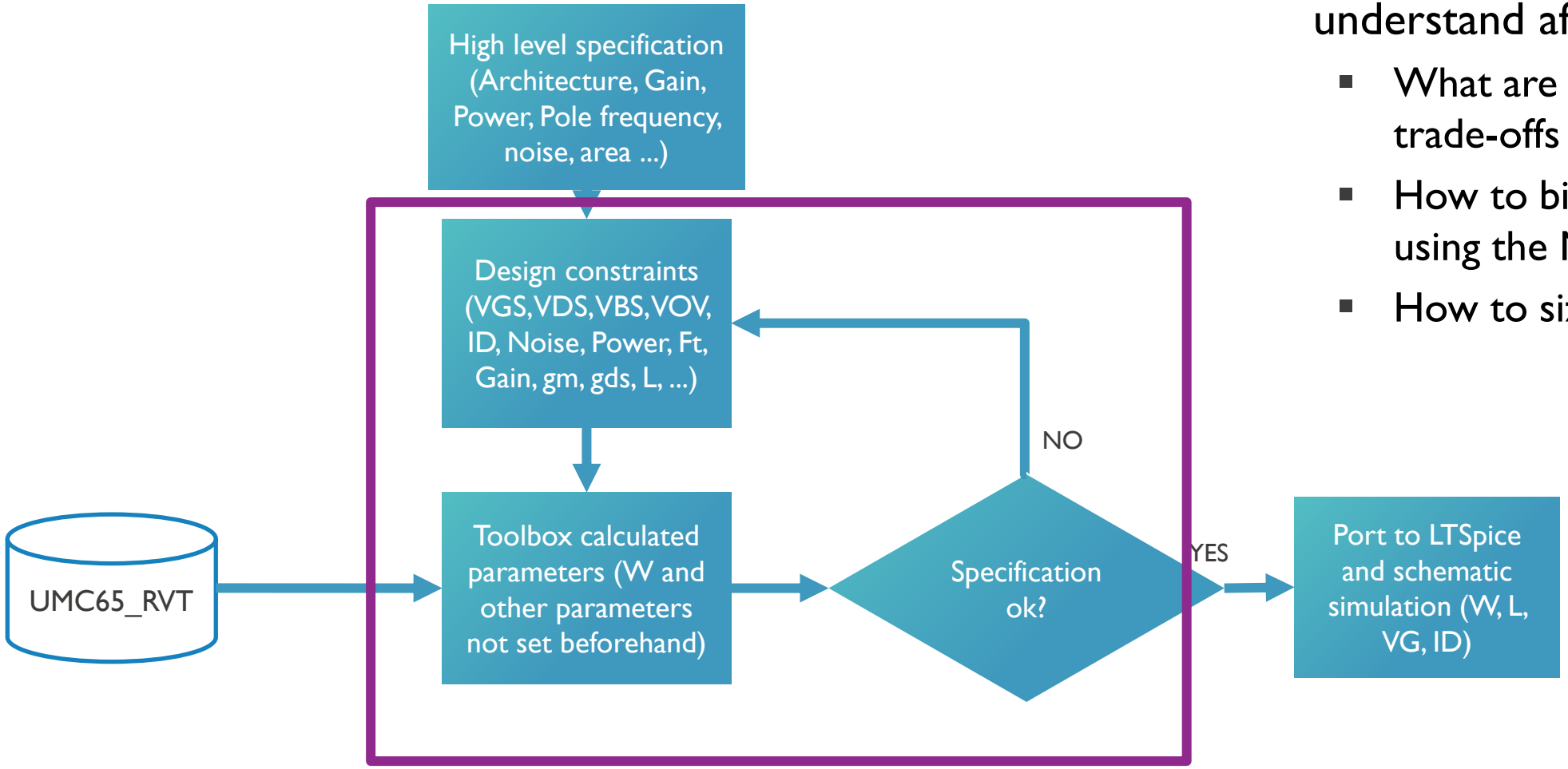
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HOW DOES THE DESIGN PROCESS WORK



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HOW DOES THE DESIGN PROCESS WORK



- What do you need to understand after this class?
  - What are the basic MOS trade-offs for design
  - How to bias a transistor using the MATLAB toolbox
  - How to size a transistor

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BUILDING SOME INTUITON

- Do exercises 1 and 2
  - ETA: 10 minutes
- Hints
  - Strong inversion: $I_{ds} \approx C_1 \cdot V_{ov}^2$ (or with velocity saturation $I_{ds} \approx C_2 \cdot V_{ov}$)
  - Weak inversion: $I_{ds} \approx C_3 \cdot \mathrm{e}^{\mathrm{Vov}} \rightarrow \ln I_{ds} \approx C_4 \cdot V_{ov} + C_5$
  - For exercise 2: there are many correct ways to obtain $1\ \mathrm{mS}$

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USING THE MATLAB TOOLBOX

BOILERPLATE **Provided to you**

```
%% Adding paths + Loading MOS tables
addpath(genpath('circuitDesign'));
addpath(genpath('functions'));
addpath(genpath('models'));

w = warning ('off','all'); % Turn off warnings

clear;
close all;
clc;

load ('UMC65_RVT.mat');

%% Initialize everything
designkitName    = 'umc65';
circuitTitle     = 'Analog Design - Session 1';

%Declaration of the circuit components
elementList.nmos = {'Mn1','Mn2','Mn3','Mn4'};
elementList.pmos = {};

spec.VDD         = 1.1;
choice.maxFingerWidth = 10e-6;
choice.minFingerWidth = 200e-9;
simulator        ='spectre';
simulFile        = 0;
simulSkelFile    = 0;
analog = cirInit('analog', circuitTitle, 'top', elementList, spec , choice,...
    designkitName, NRVT, PRVT, simulator, simulFile, simulSkelFile);

analog           = cirCheckInChoice(analog, choice);
```

Loading functions and .mat file

Defining the transistors used in the circuit (Mn***X*** is a struct with the transistor parameters as fields)

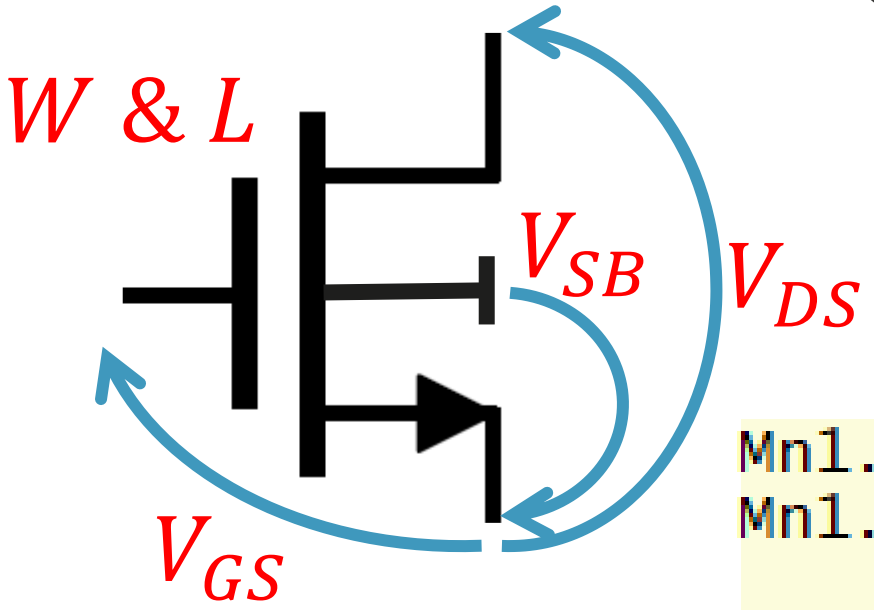
Defining technology constraints

Only change transistor list

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USING THE MATLAB TOOLBOX
SETTING THE DEVICE OPERATING POINT

1. Set values



All Required!
(S.I. Units)

```
Mn1.w = 10e-6;
Mn1.lg = 500e-9;

Mn1.vsb = 0;
Mn1.vgs = 0.4;
Mn1.vds = 0.55;
```

2. Cut fingers

```
Mn1 = mosNfingers(Mn1);
```

3. Calculate operating point

```
Mn1 = mosOpValues(Mn1);
```


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USING THE MATLAB TOOLBOX

GETTING V_{GS} FROM V_{OV}

$$V_{GS} = V_{OV} + V_{th}$$

NRVT: NMOS
PRVT: PMOS

V_{GS}
Doesn't Influence V_{th}

V_{SB}

```
Mn1.vth = tableValueWref('vth', NRVT, Mn1.lg, 0, Mn1.vds, Mn1.vsb);
```

Length

V_{DS}

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USING THE MATLAB TOOLBOX

GETTING W

If you know I_{DS}

```
Mn1.w    = mosWidth('ids', Mn1.ids, Mn1);
```

If you know g_m

```
Mn1.w    = mosWidth('gm', Mn1.gm, Mn1);
```


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PLOTTING THE MOS IDS CURVES
AGAINST VGS AND VDS

```
%% Circuit
disp('        Vd            ');
disp('        |             ');
disp('  Vg---Mnx            ');
disp('        |             ');
disp('        Vs            ');

fprintf('\n--- First Exercise: Designing transistor from scratch ---\n');
```

- **session1_part1.m**
- Fixed parameters: $W = 10\mu m, L = 65nm$
- Sweep parameters: VGS, VDS
- Calculated parameters: IDS (to be plot), VTH (byproduct of bias point) and other Operating point parameters (gm, gds, ...)

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WRITING THE CODE TOGETHER

USEFUL FUNCTIONS

tableValueWref

```
tableValueWref retrieving the value of an intrinsic MOS parameter for
the reference width, directly from a given table.

   RESULT = mosIntValueWref(PARAM, TABLE, LENGTH, VGS, VDS, VSB) returns
   the value of the MOS operating point parameter PARAM
   (specified as a string) for a given TABLE.
   PARAM must be exist as a field of the given TABLE.
   Possible names for PARAM can be found by running tableDisplay(TABLE).

   EXAMPLE :

      id = tableValueWref('ids', N, 0.18e-6, 0.7, vdd/2, 0)
```

- To extract the VTH value (weak dependency with MOS W)

mosNfingers

```
mosNfingers computation of the number of fingers for a MOS transistor.
```

- To slice the Width according to the maximum desired finger width

mosOpValues

```
mosOpValues computation of all operating parameters of a MOS transistor

   MOS = mosOpValues(MOS) computes the value of intrinsic and extrinsic
   operating point parameters (in S.I. units) of a given transistor MOS. The
   datastructure of the transistor which is specified as an argument is also
   returned. However, after this function has returned, all fields of the
   transistor related to intrinsic and extrinsic operating point parameters
   have been added to the datastructure of the transistor. Existing extrinsic
   geometry parameters are NOT recomputed. Also, the
   values of vgb, vgd and vdb are computed (or updated) from vgs, vds and vsb.

   EXAMPLE :

      Mn1 = mosOpValues(Mn1)
```

- To calculate all the other MOS parameters, in this case it is mainly Mn.ids

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ANALYSIS

- IDS graph
  - Reset to X-Z view: which regions can you see?
  - Reset to Y-Z view: which regions can you see?
- MOS saturation graph
  - Reset to Y-X view: can you discuss about the trends in Vdsat (Vds after which the transistor is in saturation)?

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BUILDING SOME INTUITION

- Exercise 3 and 4
  - ETA: 15 minutes
- In order to match results with the solution, use:
  - Weak inversion: $V_{OV} = -0.05\text{V}$
  - Moderate inversion: $V_{OV} = 0.1\text{V}$
  - Strong inversion: $V_{OV} = 0.5\text{V}$

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PLOTTING DEPENDENCY TO CHANNEL LENGTH

EXTRACTING SMALL SIGNAL PARAMETERS

```
%% Circuit
disp('     spec.VDD/2        ');
disp('        |              ');
disp('        |              ');
disp('  Vg---Mnx             ');
disp('        |              ');
disp('        Vs             ');

fprintf('\n--- First Exercise: Designing transistor from scratch ---\n');
```

- **session1_part2.m**
- Fixed parameters: VDS, W/L
- Sweep parameters: L, VGS
- Calculated parameters: g_m, g_{ds}, g_m/I_{DS} (gain efficiency), g_m/g_{ds} (intrinsic gain)

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ANALYSIS

- From an intrinsic gain perspective, which one is better?
  - Short channel or long channel?
  - Weak inversion or strong inversion?
- From a (gain) efficiency perspective, which one is better?
  - Short channel or long channel?
  - Weak inversion or strong inversion?

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HOMEWORK

- Extract the cutoff frequency (Mn.ft) value and plot alongside the previous graphs
  - For higher ft which length and inversion region are preferable?
- Extract the width calculated within the previous code and plot alongside the previous graphs
  - Which length and inversion level are preferable given a minimized area specification?

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RESISTIVE LOAD COMMON SOURCE STAGE

```
%% EX1: Circuit
disp('       VDD           ');
disp('        |            ');
disp('        RL           ');
disp('        |----+-OUT   ');
disp('  IN---Mn1   |       ');
disp('        |    CL      ');
disp('        |    |       ');
disp('       GND  GND      ');
```

- **session1_part3.m**
- Specification: Gain (>18dB), GBW (>100MHz) and Load (1pF) (given by system level decisions)
- Design choices: L, inversion region (VOV = VGS-VTH)
- Useful equations: $gm = 2\pi \times GBW \times C_{load}$ (note that gm turns to be a specification), $A_v = \frac{g_m}{g_{ds}+1/R_L}$

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WRITING THE MATLAB CODE

1. GBW = 100MHz, CL = 1pF, Gain > 18dB, Width < 1mm
2. Calculate target gm from CL and GBW
3. Set your design choices (Mn.lg, Mn.vov)
4. Set the terminal voltages (Mn.vds, Mn.vsb, VD = VDD/2)
5. Extract VTH
6. Set gate voltage (Mn.vgs = VOV+VTH)
7. Calculate width to realize given gm (Mn.w = mosWidth('gm', spec.gm, Mn))
8. Extract IDS from mosOpValues
9. Calculate RL from IDS and VD
10. Calculate gain (equation in previous slide)
    - If the specification was met, the design is ready to go to LTspice, if not go back to your design choices (lg and VOV) and start again reasoning towards which direction you want to go (long x short channel; weak x strong inversion)

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BUILDING SOME INTUITION

- Exercise 5
  - ETA: 15 minutes

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PMOS LOAD COMMON SOURCE STAGE

```
%% EX2: Circuit
disp('       VDD           ');
disp('        |            ');
disp('       Mp2           ');
disp('        |----+-OUT    ');
disp('  IN---Mn2   |        ');
disp('        |    CL       ');
disp('        |    |        ');
disp('       GND  GND       ');
```

- Now there are two choices of L and VOV. What is the reasoning behind the first guess for length? And for the first guess of inversion level?
- Gain definition: $A_v = \frac{g_m^N}{g_{ds}^N + g_{ds}^P}$

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WRITING THE MATLAB CODE

1. GBW = 100MHz, CL = 1pF, **Gain > 26dB**, Width < 1mm
2. NMOS design has the same process as the case before
3. Sizing the PMOS load has some hard constraints
    1. Mp.vds + Mn.vds = spec.vdd (forced by supply)
    2. Mp.ids = Mn.ids (to provide a load functionality)
4. PMOS width is calculated based on its required IDS (Mp.ids = mosWidth('ids', Mp.ids, Mp))

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